EXISTENCE, UNIQUENESS AND FINITENESS OF THE SCHEME THEORETIC SUPPORT OF A FINITE TYPE QUASI-COHERENT MODULE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-morphisms/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 728-744 and proof span 746-769 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof ends with the sentence 'We omit the proof.' after naming the algebra fact that the annihilator of a finite module commutes with localization at an element. That sentence is kept verbatim; what it leaves to the reader is that single elementary statement, which is displayed in the counted span, while the existence, uniqueness, minimality, finiteness and support assertions are all reduced inside the span.

Editorial note. 5 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The same-chapter equivalence between quasi-coherent modules on a closed subscheme and quasi-coherent modules on the ambient scheme annihilated by the ideal sheaf, which gives the uniqueness in the form used here. Cited at proof lines 964-967 and again at 969-970. Converts the condition that the module is pushed forward from a closed subscheme into a statement about the annihilator, and gives uniqueness of the pushed forward module.
- The same-chapter bijection between closed subschemes of an affine scheme and ideals of the coordinate ring, used to see that the annihilator rule defines a closed subscheme. Cited at proof lines 979-980. Reduces the existence of the smallest closed subscheme to showing that the annihilator rule gives a quasi-coherent sheaf of ideals.
- Over an affine scheme a finite type quasi-coherent module is the tilde of a finite module, and the image of the support of the module is the closed subset defined by the annihilator. Cited at proof lines 971-978. Gives the finiteness of the pushed forward module and the identification of supports.

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0560: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6693–6693. Same-chapter reference used in the retained author text at source line(s) 756. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 00L2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 772–772. Same-chapter reference used in the retained author text at source line(s) 705, 759, 11631, 11640, 11679, 11688. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01QQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2275–2275. Same-chapter reference used in the retained author text at source line(s) 268, 511, 764, 815. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01QY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2283–2283. Same-chapter reference used in the retained author text at source line(s) 578, 749, 782, 6149, 6758. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01PB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2224–2224. Same-chapter reference used in the retained author text at source line(s) 703, 754, 4224, 5201, 5286, 6703, 10562, 11247. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let $\mathcal{F}$ be a finite type quasi-coherent module on a scheme X. There exists a smallest closed subscheme $i : Z \to X$ such that there exists a quasi-coherent $\mathcal{O}_Z$-module $\mathcal{G}$ with $i_*\mathcal{G} \cong \mathcal{F}$. Moreover:*

1. *If $\mathrm{Spec}(A) \subset X$ is any affine open, and $\mathcal{F}|_{\mathrm{Spec}(A)} = \widetilde{M}$ then $Z \cap \mathrm{Spec}(A) = \mathrm{Spec}(A/I)$ where $I = Ann_A(M)$.*
2. *The quasi-coherent sheaf $\mathcal{G}$ is unique up to unique isomorphism.*
3. *The quasi-coherent sheaf $\mathcal{G}$ is of finite type.*
4. *The support of $\mathcal{G}$ and of $\mathcal{F}$ is Z.*

Proof. Suppose that $i' : Z' \to X$ is a closed subscheme which satisfies the description on open affines from the lemma. Then by Lemma 01QY we see that $\mathcal{F} \cong i'_*\mathcal{G}'$ for some unique quasi-coherent sheaf $\mathcal{G}'$ on Z'. Furthermore, it is clear that Z' is the smallest closed subscheme with this property (by the same lemma). Finally, using Properties, Lemma 01PB and Algebra, Lemma 0560 it follows that $\mathcal{G}'$ is of finite type. We have $\mathrm{Supp}(\mathcal{G}') = Z$ by Algebra, Lemma 00L2. Hence, in order to prove the lemma it suffices to show that the characterization in (1) actually does define a closed subscheme. And, in order to do this it suffices to prove that the given rule produces a quasi-coherent sheaf of ideals, see Lemma 01QQ. This comes down to the following algebra fact: If A is a ring, $f \in A$, and M is a finite A-module, then $\mathrm{Ann}_A(M)_f = \mathrm{Ann}_{A_f}(M_f)$. We omit the proof. □